

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQs

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – MCQs
Weightage:	10% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	Ruthika Reddy C	Reg. No.:	192421138
Section / Batch:	Btech IT	Date:	

Instructions to Students

- Answer all questions carefully. Each question carries 2 marks. Total: 100 marks.
- Analyze each question and select the correct answer.
- Apply concepts correctly to solve problem-based MCQs.
- Manage your time efficiently to complete all 50 questions.

Question Type	Focus Area	Questions
MCQs	Analyze and apply Divide and Conquer Algorithms for Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication.	Q1 -Q50

SECTION A – MCQs

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Online Quiz / MCQ Test					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Mark Evaluations:

No. of MCQ Questions	Correct Answers	Wrong/Not Answers	Total Marks (100)
50			

Faculty In-charge	Course Coordinator	Head of Department
Name: _____ Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS Online Programming Lab

DAA PROGRAMMING



C RUTHIKA REDDY
192421138RA

ASSESSMENT CO3-AT3-MCQ

[← Back](#)

Language:

Python

C

C++

Java

QUESTIONS

Q1

Binary Search Tree traversal complexity
2 marks

**Q2**

Time complexity of partitioning in Quick Sort
2 marks

**Q3**

Strassen's multiplication reduction
2 marks

**Q4**

Structure unsuitable for Binary Search
2 marks

**Q5**

Binary Search precondition
2 marks

**Q6**

Merge Sort base case for recursion
2 marks

**Q7**

Merge Sort preferred over Quick Sort for linked lists

**Q1**

Binary Search Tree traversal complexity

2

marks

Binary Search tree traversal visits all nodes sequentially. Analyze the time complexity of traversal operations.

A. $O(n)$ **B.** $O(\log n)$ **C.** $O(n \log n)$ **D.** $O(n^2)$

Save Answer

Next →

2 marks

Q8



Number of parts array divided
in Merge Sort

2 marks

Q9



Binary Search mid index
formula

2 marks

Q10



Merge Sort time complexity in
all cases

2 marks

Q11



Divide and Conquer steps

2 marks

Q12



Condition where Binary Search
fails

2 marks

Q13



Merge Sort recurrence relation

2 marks

Q14



Type of arrays merged in Merge
Sort

2 marks

Q15



Quick Sort on sorted array with
first pivot

2 marks

Q16



Quick Sort worst-case pivot
condition

2 marks

Q17



Element used for comparison in
Binary Search

2 marks

Q18



Binary Search comparisons for
 $n=1024$
2 marks

Q19

Binary Search complexity for
unsuccessful search
2 marks

Q20

Reason Quick Sort is called in-
place
2 marks

Q21

Reason Merge Sort is not in-
place
2 marks

Q22

Property ensuring Merge Sort
stability
2 marks

Q23

Quick Sort best performance
condition
2 marks

Q24

Paradigm followed by Merge
Sort
2 marks

Q25

Binary Search best-case
complexity
2 marks

Q26

Operation minimized by
Strassen's algorithm
2 marks

Q27

Name of Merge Sort's
combining step
2 marks

Q28

Technique used by Strassen's
algorithm

2 marks

Q29

Condition causing Quick Sort inefficiency

2 marks

Q30

Worst-case comparisons in Binary Search

2 marks

Q31Binary Search comparisons for $n=1000$

2 marks

Q32

Condition where Binary Search is most effective

2 marks

Q33

Strassen's algorithm multiplication reduction (2)

2 marks

Q34

Merge Sort auxiliary space complexity

2 marks

Q35

Average recursion depth of Quick Sort

2 marks

Q36

Binary Search recurrence relation

2 marks

Q37

Strassen's algorithm complexity

2 marks

Q38

Strassen's algorithm efficiency condition

2 marks

Q39

Reason Merge Sort preferred
for external sorting

2 marks

Q40



Main advantage of Quick Sort

2 marks

Q41



Merge Sort recursion tree
height

2 marks

Q42



Quick Sort worst case with poor
pivot

2 marks

Q43



Hybrid sort switching to
Insertion Sort

2 marks

Q44



Time complexity of recursive
divide-and-conquer algorithm

2 marks

Q45



Number of parts data divided
in Binary Search

2 marks

Q46



Stable sorting algorithm

2 marks

Q47



Strassen's algorithm reduced
operation

2 marks

Q48



Reason Quick Sort is unstable

2 marks

Q49



Quick Sort average time
complexity

2 marks

Q50



Pivot position after partitioning
2 marks

50/50 answered

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